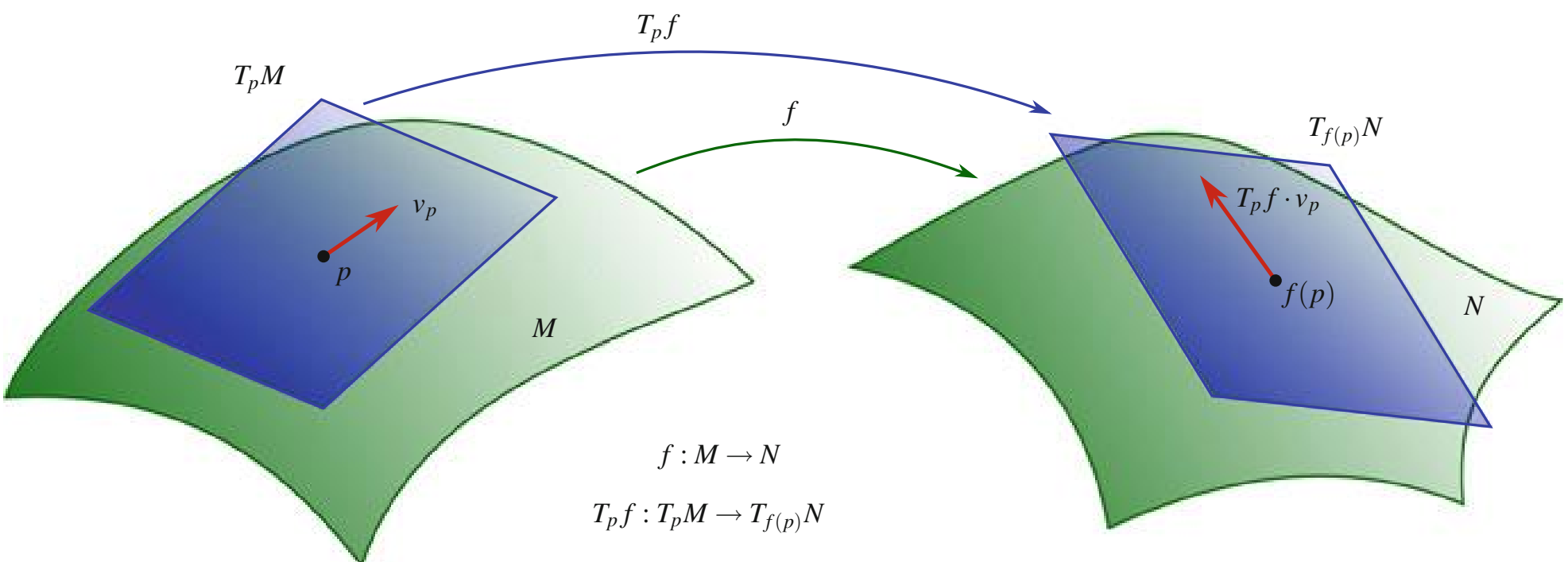

T_pf
T_pM
f
$T_{f(p)}N$
v_p
$T_pf \cdot v_p$
p
$f(p)$
M
N
$f : M \to N$
$T_pf : T_pM \to T_{f(p)}N$